

Q6.

```

DECLARE VType, Member, Disabled, Available: CHAR
DECLARE CSOC, RSOC, Time: INTEGER
DECLARE PDuration, RCharge, BRate, OPercentage: REAL
DECLARE RCost, DAmnt, FCost: REAL
DECLARE TDiscount, TPay: REAL
DECLARE Prior, Times, Out: STRING
DECLARE LCost, PCost, BPark: INTEGER
INPUT VType
INPUT CSOC
INPUT RSOC
INPUT PDuration
INPUT Time
INPUT Member
INPUT Disabled
INPUT Available

```

```

LCost ← 0, PCost ← 0, DAmnt ← 0,
RCharge ← 0, Prior ← "None"
Times ← "None"
Out ← " "

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IF Available = "N" THEN

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    IF VType = 'H' THEN

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        OUTPUT "Charging Unavailable - Parking only"
    ELSE

```

```

        OUTPUT "No charging slot Available"
    ENDIF

```

```
CALL P-CALL
ENDIF
```

```
IF VType = 'H' AND (SOC >= 40) THEN
    OUTPUT "Vehicle does not qualify for EV
    charging"
```

```
CALL P-CALL
ENDIF
```

```
RCharge ← RSOC - Curr SOC
```

```
IF RCharge ≤ 0 THEN
```

```
    OUTPUT "No charging required"
```

```
    CALL P-CALL
```

```
ENDIF
```

```
IF CSOC ≤ 15 AND RSOC ≥ 80 THEN
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```
    Prior ← "Emergency charging Priority"
```

```
ELSE IF Disables = 'y' OR (Member = 'y' AND
    (SOC ≤ 30)) THEN
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```
    Prior ← "Priority charging"
```

```
ELSE
```

```
    Prior ← "Normal charging"
```

```
ENDIF
```

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DPercentage ← 0
IF Time ≥ 17 AND Time < 22 THEN
    Times ← "Peak"
    BRate ← 50
    IF Member = 'Y' AND Priority
        Prior <> "Emergency charging Priority"
    THEN
        DPercentage ← 0.1
    ENDIF
ELSE
    Times ← "Off-Peak"
    BRate ← 35
    IF Member = 'Y' AND
        Pri.Prior <> "Emergency charging
        Priority" THEN
        THEN
            DPercentage ← 0.2
        ENDIF
    ENDIF
ENDIF

```

```

RCost ← RCharges * BRate
DAmt ← RCost * DPercentage
CCost ← RCost - DAmt
TDiscount ← TDiscount + DAmt

```

PROCEDURE CALCULATION

IF PDuration ≤ 2 THEN

BPark $\leftarrow 200$

ELSE IF PDuration ≤ 5 THEN

BPark $\leftarrow 400$

ELSE

BPark $\leftarrow 700$

ENDIF

IF Disabled = 'Y' THEN

PCost $\leftarrow 0$

TDiscount $\leftarrow$ TDiscount + BPark

ELSE IF Member = 'Y' THEN

DAmnt $\leftarrow$ BPark * 0.2

PCost $\leftarrow$ BPark - DAmnt

TDiscount $\leftarrow$ TDiscount + DAmnt

ELSE

PCost $\leftarrow$ Base_Park.

ENDIF

TPay $\leftarrow$ Cost + PCost

OUTPUT " — System Report — "

OUTPUT "Vehicle Type:", VType

OUTPUT "Current Battery:", CSOC, "%"

OUTPUT "Required Battery:", RSOC, "%"

OUTPUT "Priority:", PId.

OUTPUT "Time Status:", Time S.

```

OUTPUT " ——— Cost ——— "
OUTPUT "Charging cost Rs: ", Lcost
OUTPUT "Parking cost Rs: ", Pcost
OUTPUT "Discounts Applied Rs: ",
      TDiscount
OUTPUT "Final Amount Rs: ",
      TPay
OUTPUT "Note: ", Note
IF PDuration > 8 THEN
  out ← "Long-stay Warning:
        Please Relocate your
        vehicle After charging"
ELSE
  out ← "Standard parking Duration"
ENDIF

OUTPUT "Note: ", out.
ENDPROCEDURE

```